# P101 · Unit 4

## Interactions, diagrams and motion

Recall: retrieve readily · Understand: reconstruct reasoning · Consult: look up as needed

| Use | Topic | Result and conditions |
|---|---|---|
| Recall | **System & diagram** | Select the boundary. Draw external forces with agent, recipient and direction; components replace, not supplement, a force. |
| Recall | **Newton's laws** | $\sum F_{\text{ext}} = ma$; constant mass, inertial frame. Zero resultant $\Rightarrow$ constant velocity (possibly zero). |
| Recall | **Third-law pair** | $F_{A \to B} = -F_{B \to A}$; different recipients. Internal pairs cancel in the combined sum. |
| Recall | **Weight & units** | $W = mg$; $W = mg$. $1\,\text{N} = 1\,\text{kg}\,\text{m}\,\text{s}^{-2}$; g in $\text{m}\,\text{s}^{-2}$ or $\text{N}\,\text{kg}^{-1}$. |
| Understand | **Support & cord** | Find $N$ from perpendicular motion, with $N \geq 0$ for nonadhesive contact. Taut rope pulls; $T \geq 0$. Light straight rope, no other loads: equal endpoint tension. |
| Recall | **Dry friction** | Sticking: $0 \leq f_s \leq \mu_s N$; calculate the required force, then test. Sliding: $f_k = \mu_k N$, opposite relative slip. Recheck at a stop. |
| Consult | **Fixed ramp** | Only weight + contact: $N = mg\cos\theta$; downslope weight $mg\sin\theta$. Rest possible if $\tan\theta \leq \mu_s$, for $0 \leq \theta < 90°$. |
| Recall | **Constant acceleration** | $s = x - x_0$, $t$ elapsed, $u$ initial signed velocity. $v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$. One fixed axis, constant $a$ throughout. |
| Understand | **Areas & intervals** | $s = \frac{1}{2}(u + v)t$ for constant $a$: signed velocity–time area. Distance: add magnitudes across turns. At $a = 0$: $v = u$, $s = ut$. Stop when the force regime changes. |


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## Projectile motion & Python

| Use | Topic | Result and conditions |
|---|---|---|
| Recall | **Projectile components** | Uniform gravity, no drag; right/up axes: $u_x = u\cos\theta$, $u_y = u\sin\theta$. $x = x_0 + u_x t$, $y = h + u_y t - \frac{1}{2}gt^2$; $v_x = u_x$, $v_y = u_y - gt$. One shared time. |
| Consult | **Floor event** | For $h \geq 0$, $u_y \geq 0$, $g > 0$: $t_f = (u_y + \sqrt{u_y^2 + 2gh}) / g$. $R = u_x t_f$ with $u_x \geq 0$. At $h = u_y = 0$: no airborne interval. Earlier obstacles can end flight. |
| Understand | **Top & equal height** | For $u_y > 0$: $t_{\text{top}} = u_y / g$, $y_{\max} = h + u_y^2 / (2g)$. At top $v_y = 0$, $a_y = -g$. Equal-height later return only: $t_f = 2u_y / g$. |

### Python lookup

Imports: `import numpy as np`; `import matplotlib.pyplot as plt`. Finite SI inputs; h ≥ 0, uy ≥ 0, g > 0; floor y = 0. Numerical tolerance is not measurement uncertainty.

**Consult · components in SI**

```
theta = np.deg2rad(angle_deg)
ux = speed * np.cos(theta)
uy = speed * np.sin(theta)
```

**Consult · sampled algebraic path**

```
tf = (uy + np.sqrt(uy**2 + 2*g*h)) / g
t = np.linspace(0.0, tf, 101)
x = ux*t
y = h + uy*t - 0.5*g*t**2
```

**Consult · plot & checks**

```
fig, ax = plt.subplots()
ax.plot(x, y)
ax.set_xlabel("x / m"); ax.set_ylabel("y / m")
ax.set_aspect("equal", adjustable="box")
fig.savefig("path.png", dpi=160)
print(abs(y[-1]) < 1e-10)
```

Understand: benchmark h = 4.9 m, uy = 0, ux = 6 m/s, g = 9.8 m/s² → tf = 1 s, x = 6 m. Check initial state, floor event, known angle, units and clean rerun. No measured data → no physical validation.


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